

Skin Disease Classification Model Comparison Report

Phase 4 - Baseline Model Performance

Model	Train Acc	Val Acc	Test Acc
Basic CNN	39.39%	34.29%	31.80%
Deep CNN	46.39%	44.54%	43.51%
CNN + BatchNorm	52.08%	51.00%	46.71%
CNN + Dropout	50.01%	49.60%	49.04%
MobileNetV2	55.15%	53.06%	50.38%
DenseNet121	54.17%	56.06%	55.89%
EfficientNetB0	58.35%	59.92%	59.02%
ResNet50	25.86%	30.49%	31.07%

Phase 7 - Final Model Performance

Model	Accuracy	Precision	Recall	F1
Basic CNN	53.76%	73.58%	53.76%	59.03%
Deep CNN	66.93%	44.80%	66.93%	53.67%
CNN + BatchNorm	47.44%	65.76%	47.44%	53.28%
CNN + Dropout	52.56%	74.03%	52.56%	57.24%
MobileNetV2	63.67%	70.09%	63.67%	66.26%
EfficientNetB0	62.93%	44.80%	62.93%	53.67%
ResNet50	66.93%	44.80%	66.93%	53.67%
DenseNet121	64.60%	74.43%	64.60%	68.30%

Final Selected Model: ResNet50

The final model selected for deployment is ResNet50. The model provides disease classification with explainability using Grad-CAM and Saliency Map techniques.